

```

<ion-navbar>
  <ion-title>Movie Details</ion-title>
</ion-navbar>

<ion-content>
  <div ng-if="movie" class="selection">
    <ion-card>
      <ion-item>
        <ion-avatar item-left image-large>
          
        </ion-avatar>
        <ion-item-content class="movie-title-data">
          <h1>{{movie.title}}</h1>
          <p>{{movie.release_date}}</p>
        </ion-item-content>
      </ion-item>
    </div>
  </ion-content>

```

Figure 3: Source code snippet in the Ionic framework

its striking features is the presence of AngularJS at its core and support for the Syntactically Awesome Style Sheets (SASS) CSS extension. To interact with mobile hardware and other device capabilities, it makes use of Apache Cordova. To learn more about this framework, you can visit <https://ionicframework.com/>.

2. **Onsen UI:** This is a comparatively new framework that is moving up on the development scene very fast. It is also open source and licensed under Apache, and has two versions -- Onsen UI and Onsen UI2. It is somewhat similar to the Ionic framework in terms of how it works with AngularJS, but Onsen UI2 is really useful because it is platform agnostic. If you are comfortable with jQuery, you can use jQuery too. It is a very flexible and performance driven framework. For more information, you can go to <https://onsen.io/>.
3. **jQuery Mobile:** This is one of the oldest frameworks still in use, having been first released in 2010. It is an HTML5 mark-up based framework. This means that most of the work is done through HTML5. Apps designed using this framework will run on anything that supports the HTML5 standard. It has a lot of documentation and online support available. Though it has some performance concerns, it still is the most widely used after the Ionic framework. You can read more about it on <https://jquerymobile.com/>.
4. **Sencha Touch:** This is one of the oldest licensed tools and is mostly used by enterprises for building cross-platform, high-performance mobile apps quickly and effectively. It uses Apache Cordova and PhoneGap to support device hardware. It comes packaged with ready-to-use widgets that can work on all popular platforms like iOS, Android, etc. You can visit <https://www.sencha.com/> for more information on Sencha Touch.
5. **Intel XDK:** This is new in the hybrid mobile app framework space but due to its ordered workflow and ready-to-use templates, it is becoming the favoured choice for developing cross-platform apps. It contains all the tools required, from the development to the testing of HTML5 hybrid apps.
6. **Kendo UI:** This is supported by Telerik. It makes

```

<body>
  <div data-role="page" id="home">
    <div data-theme="a" data-role="header">
      <h3>Movie List</h3>
    </div>
    <div data-role="content">
      <div class="example-wrapper" data-is-scroll>
        <ul data-role="listview" id="movie-list" data-theme="a">
          </ul>
        </div>
      </div>
    <div data-theme="a" data-role="footer">
      <h3>Copyright 2015</h3>
    </div>
  </div>

```

Figure 4: Source code in the jQuery Mobile framework with the 'data-role' and 'data-theme' members

- use of jQuery for development. It can also interact with AngularJS and the Bootstrap UI framework. It is available as an open source as well as a commercial version, though the former has fewer features. Do visit <http://www.telerik.com/kendo-ui> for more information on Kendo UI.
7. **Framework 7:** This is a totally free and open source framework licensed under MIT and used only for developing iOS and Android apps. It is not a cross-platform framework but due to its look and feel, seems like a native app. It is being widely used nowadays. You can find a number of apps on the Apple App store and Google Play store that have been developed using Framework 7. If you wish to read more about it, visit <https://framework7.io/>.
 8. **Famo.us:** This framework is mostly used when you have to deal with high-end rendering and graphics for games. It is an open source framework and can provide hybrid apps with the look of native apps. Visit <https://famous.co/> for more information.

Making the right choice

Mobility is definitely the future of technology. With the entry of so many technologies and frameworks, the issue is of making the right choice when developing mobile applications. This choice depends on various aspects such as the mobile platforms to target, the distribution of the applications, cost considerations, the technical competence of the development team, mobile hardware capabilities, security, the target audience, the user experience and the time to market. Watching how HTML5 has advanced over the past few years, one can definitely speculate that the future is bright for Web developers who want to break into the world of mobile application development. **END** 

By: Vinayak Vaid

The author works as an automation engineer at Infosys Limited, Pune. He has worked on different testing technologies and automation tools like QTP, Selenium and Coded UI. He can be contacted at vinayakvaid91@gmail.com.